

1.

Answer to the question no. 1

Ans: Quantum computing threatens to break RSA and ECC. Shor's algorithm can efficiently solve the mathematical problems (integer factorization and discrete logarithms) they rely on.

Proposed replacements (Post-Quantum Cryptography):

Lattice-based (Kyber, Dilithium): Hardness based on shortest vector problems in high-dimensional lattices.

Hash-based (SPHINCS+): Relies on the security of cryptographic hashes.

Code-based (McEliece): Based on the difficulty of decoding general linear codes.

These resist quantum cryptanalysis because unlike factoring, these problems are currently believed to be unsolvable in polynomial time by quantum computers.

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2.

Answer to the question no. 2

Ans: Here is a minimal implementation using a linear congruential Generator (LCG) approach seeded by the system state:

```
import time
```

```
import os
```

```
class CustomPRNG:
```

```
    def __init__(self, modulus=100000):
```

```
        self.modulus = modulus
```

```
        self.state = int(time.time_ns()) ^ os.getpid()
```

```
    def random(self):
```

```
        self.state = (self.state * 1103515245 + 12345 +  
                      os.getpid()) % self.modulus
```

```
        return self.state
```

```
# Example usage
```

```
prng = CustomPRNG(modulus=1000)
```

```
for _ in range(5):
```

```
    print(prng.random())
```


3

Answer to the question no. 3

Ans: Traditional ciphers (Caesar, Vigenere) are character-based, manual, and easily broken by frequency analysis due to short keys and predictable pattern.

Modern ciphers (DES, AES) are bit-based, highly complex and mathematically robust.

Comparison:

Key length: Traditional (Tiny/variable) vs DES (56-bit)
vs AES (128-256 bit)

Speed: Traditional is slow manual; modern is extremely fast.

Security: Traditional is insecure; DES is obsolete (brute-forceable); AES is the current global gold standard.

4.

Ans: Action Definition: The action is defined by

$$\sigma \{i, j\} = \{\sigma(i), \sigma(j)\}$$

for any permutation $\sigma \in S_4$

Well defined: Since σ is a bijection, $\sigma(i) \neq \sigma(j)$ if any $i \neq j$. Thus the image is always a 2-element subset of $\{1, 2, 3, 4\}$

Orbit size: The set of 2-element subsets is

$$X = \{\{1, 2\}, \{1, 3\}, \{1, 4\}, \{2, 3\}, \{2, 4\}, \{3, 4\}\}.$$

Since S_4 can map any pair to any other pair, the action is transitive. Orbit size $|O_{\{1, 2\}}| = 6$.

Stabilizer size:

$$|S_4| = |\text{orbit}| \times |\text{stabilizer}|$$

$$24 = 6 \times |\text{stabilizer}|$$

$$|\text{stabilizer}| = |G_{\{1, 2\}}| = 4$$

The elements are: id , (12) , (34) and $((12)(34))$

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5.

Ans:

i) Multiplication binop: The non-zero elements are $G^* = \{1, \alpha, \alpha+1\}$

closure: $\alpha^2 = \alpha+1$; $\alpha(\alpha+1) = 1$

identity: 1 exists

inverses: Every element has a partner that multiplies to 1 ($\alpha(\alpha+1) = 1$).

G^* satisfies all group axioms.

ii) Cyclicity Yes, it is cyclic. The power of

α are: $\alpha^1 = \alpha$, $\alpha^2 = \alpha+1$ and $\alpha^3 = 1$.

Since α generates all non-zero elements the group is cyclic.

Ans: Normal Subgroup proof: Let Z be the set of scalar matrices λI (where $\lambda \neq 0$). For any $A \in GL(2, \mathbb{R})$ and $S = \lambda I \in Z$: $AS A^{-1} = A(\lambda I)A^{-1} = \lambda(A I A^{-1}) = \lambda I$

Since, $AS A^{-1} \in Z$; Z is a normal subgroup.

Factor Group: The factor group is $GL(2, \mathbb{R})/Z$ which is the Projective General Linear Group, $PGL(2, \mathbb{R})$

Interpretation: $PGL(2, \mathbb{R})$ represent transformation where scaling does not matter. Geometrically, it describes the projective transformations of the real projective line $(\mathbb{R}P^1)$, mapping lines through the origin to other lines.

7. Ans: Protocol: Two parties, Alice and Bob, agree on a large prime p and a generator g . Alice sends $g^a \pmod{p}$ and Bob sends $g^b \pmod{p}$. Both compute the shared secret $K = g^{ab} \pmod{p}$.

Security Basis: It relies on the Discrete logarithm Problem (DLP); It is computationally infeasible to find a or b from g^a or g^b given a large enough p .

Attacks & Risks:

Man-in-the middle (MitM): The protocol is vulnerable because it lacks authentication.

Small Prime Modulus: If p is too small, attackers can solve the DLP using algorithms like the Number Field sieve or brute force.

8.

Ans: Proof: Let H and K be subgroup of G .

To prove $H \cap K$ is a subgroup, we check the three axioms.

1. Identity: Since $e \in H$ and $e \in K$ then $e \in H \cap K$

2. Closure: If $a, b \in H \cap K$, then

$a, b \in H$ and $a, b \in K$ and H, K are subgroups, $ab \in H$ and $ab \in K$ so, $ab \in H \cap K$

3. Inverse: If $a \in H \cap K$, then $a \in H$ and $a \in K$. Since H and K are subgroups, $a^{-1} \in H$ and $a^{-1} \in K$, so,

$a^{-1} \in H \cap K$

Example: In the group integer $(\mathbb{Z}, +)$: Let $H = 2\mathbb{Z}$ and $K = 3\mathbb{Z}$. The intersection $H \cap K = 6\mathbb{Z}$. Which is also a subgroup of $\mathbb{Z}$.

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Ans: Commutativity: In $\mathbb{Z}_n$, multiplication is defined as $[a][b] = [ab \pmod n]$. Since integer multiplication is commutative ($ab = ba$), it follows that $[a][b] = [b][a]$. Thus, $\mathbb{Z}_n$ is a commutative ring.

Zero Divisors: $\mathbb{Z}_n$ has zero divisors if and only if n is composite. Example: In $\mathbb{Z}_6$, $[2] \cdot [3] = [6] = [0]$. Since $[2] \cdot [3] \neq [0]$, They are zero divisors.

Condition for a field: $\mathbb{Z}_n$ is a field if and only if n is a prime number. If n is prime, every non-zero element has a multiplicative inverse and there are no zero divisors.

10.

Ans: Vulnerabilities: DES is insecure primarily due to its short 56-bit key, which is vulnerable to brute-force attacks. Modern hardware can crack a DES key in hours. It is also susceptible to differential and linear cryptanalysis.

AES improvements:

Key-size: AES offers 128, 192, 256 bit keys, making brute force mathematically impossible.

Block-size: AES uses 128-bit blocks (vs DES's 64 bit) preventing data collision attacks.

Design: AES uses a substitution-permutation network, specifically designed to resist the cryptanalytic techniques that threatened DES.

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Ans: DES : Resistance relies solely on the S-boxes. The Feistel structure only modifies that the half the block per round, requiring more rounds to achieve full diffusion. While DES-S boxes were designed to minimize differential probability, the small 64 bit block size still allows attackers to track difference pattern.

ii) AES Resistance:

1. Wide Trail Strategy: Unlike DES every bit of the state is transformed every round.

2. Diffusion: ShiftRows and MixColumns ensure a single bit change spread across the entire block rapidly.

3. S-boxes - AES-S-boxes have a mathematically lower maximum differential probability making it much harder to find high-probability differential paths.

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Answer: Algorithm: Solve $ax + ny = 1$ using back-substitution from the Euclidean Algorithm. The value $x \pmod{n}$ is the inverse.

RSA Application: It is used to calculate the private key d , where $d \equiv e^{-1} \pmod{\phi(n)}$.

Efficiency: Because RSA uses huge numbers (2048-bit+); the algorithm's logarithmic speed is essential for generating keys and performing decryptions in milliseconds.

13.

Answer: i) ECB Insecurity:

Mathematical Proof: ECB encrypts blocks independently

$$C_i = E_k(P_i) \text{ If } P_i = P_j \text{ then } C_i = C_j$$

Result: ECB preserves patterns in data, identical blocks in an image remain (identical in ciphertext) leaking structural information.

ii) CBC Relations & Error Propagation

Relations:

$$\text{Encryption: } C_i = E_k(P_i \oplus C_{i-1})$$

$$\text{Decryption: } P_i = D_k(C_i) \oplus C_{i-1}$$

Error Propagation Proof: If ciphertext C_i is corrupted.

1. P_i is totally garbled (since C_i is the input to D_k)

2. P_{i+1} has bit-errors only where C_i was flipped (xor property)

3. P_{i+2} and beyond are unaffected because they depend on C_{i+1}, C_{i+2} etc, which are correct. Result: Errors are limited to exactly two blocks.

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Ans: Linearity Vulnerability

LFSRs follow a linear recurrence:

$$S_i = \sum c_j S_{i-j} \pmod{2}$$

1. Attack: An attacker with $2L$ bits of keystream can use the Berlekamp-Massey algorithm.

2. Outcome: The algorithm solves a system of linear equations in $O(L^2)$ time to recover the feedback polynomial and initial state, breaking the key.

Mitigation: Non-linear filtering, Apply a non-linear Boolean function $f(x)$ to several taps of the LFSR.

Result: This destroys the linear relationship and increases the linear complexity, making it possible for simple linear solvers to reconstruct.

15.

Ans: Definition: Perfect secrecy means:

$$P(M=m | C=c) = P(M=m)$$

Knowing the ciphertext reveals nothing about the message.

ii) OTP Proof:

1. Relation: $C = M \oplus K \Rightarrow$ for every (m, c)

there is exactly one k .

2. Probability: If k is uniform, $P(C=c | M=m) = \frac{1}{|K|}$

3. Independence: Since this value is the same for all m , M and C are independent

iii) Why Impractical: The key must be at least as long as the message. Sending a 1TB key is as hard as sending a 1TB message. Reusing the keys "on-time" only that means never destroy secrecy.

16.

Ans: To compute the sequence, let's use the parameters: $a = 3$, $c = 1$ and $m = 10$

Recurrence: $x_{n+1} = (3x_n + 1) \pmod{10}$

Calculation:

$x_0 = 7$ (seed)

$x_1 = (3 \cdot 7 + 1) \pmod{10} = 22 \pmod{10} = 2$

$x_2 = (3 \cdot 2 + 1) \pmod{10} = 7 \pmod{10} = 7$

$x_3 = (3 \cdot 7 + 1) \pmod{10} = 22 \pmod{10} = 2$

$x_4 = (3 \cdot 2 + 1) \pmod{10} = 7 \pmod{10} = 7$

Sequence: 7, 2, 7, 2, 7

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Ans: Rings in Abstract Algebra

Definition: A ring is a set with two operations $(+)$ and $(\cdot)$ where $+$ is addition and $\cdot$ is multiplication.

Distributive: $a(b+c) = ab+ac$

Example

Commutative: Integers $(\mathbb{Z})$ where $a \cdot b = b \cdot a$

Non-commutative: Square matrices, where order matters $(AB \neq BA)$

Relation to finite fields: A field is a commutative ring where every non-zero element has a multiplicative inverse.

Role in RSA: RSA works within the ring $\mathbb{Z}_n$. Encryption and Decryption are modular exponentiation operations.

18.

Ans: Part 1: RSA Encryption & Decryption

set up $(p=5, q=11, M=2)$; $n=55$, $\phi(n)=40$
let, $e=3$, then $d=27$ ($3 \times 27 \equiv 1 \pmod{40}$)

Encryption: $C = 2^3 \pmod{55} = 8$

Decryption: $M = 8^{27} \pmod{55} = 2$

Part-2: RSA Signature

setup $(p=7, q=3, H(m)=3)$:

$n=21$, $\phi(n)=12$

let $e=5$ then $d=5$ ($5 \times 5 \equiv 1 \pmod{12}$)

sign: $S = 3^5 \pmod{21} = 243 \pmod{21} = 12$

verify: $V = 12^5 \pmod{21} = 3$ (Matches $H(m)$)

Why it works

1. Authenticity

2. Integrity

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Ans: i) Verify $P(8, 10)$

$$y^2: 10^2 = 100 \equiv 8 \pmod{29}$$

$$x^3: x^3 + x + 1: 8^3 + 8 + 1 = 513 \equiv 8 \pmod{29}$$

$$\text{result: } 8 = 8$$

ii) Double $P(2P)$

$$\text{Slope } (\lambda): \frac{3(y^2 + 1)}{2x} = \frac{3(10^2 + 1)}{2(8)} = \frac{3(101)}{16} \equiv \frac{303}{16} \equiv \frac{5}{16} \equiv \frac{1}{4} \equiv 6 \pmod{29}$$

$$2. \quad 2y: 2(10) = 20 \equiv 7 \pmod{29}$$

$$3. \quad y^2: 6(7^2) - 10 = 294 - 10 = 284 \equiv 12 \pmod{29}$$

$$\text{result: } 2P = (7, 12)$$

iii) Addition $(P+Q)$

$$1. \quad \text{Slope } (\lambda): \frac{y_2 - y_1}{x_2 - x_1} = \frac{10 - 8}{9 - 3} = \frac{2}{6} \equiv \frac{1}{3} \equiv 11 \pmod{29}$$

$$2. \quad x_3: 11^2 - 3 - 9 = 121 - 12 = 109 \equiv 17 \pmod{29}$$

$$3. \quad y_3: 11(3 - 17) - 10 = -154 - 10 = -164 \equiv 20 \pmod{29}$$

$$\text{result: } P+Q = (17, 20)$$

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Ans: i) Public key G $G = 96$; calculated via scalar multiplication of $(2, 5)$ Result : $G = (3, 3)$

ii) ECDSA signing

1. compute: calculate $kG = 3G = (15, 20)$

$$r = 15 \pmod{19} = 15$$

2. compute s :

$$s = k^{-1} (H(M) + rd) \pmod{n}$$

$$s = 3^{-1} (8 + 15 \cdot 9) \pmod{19} = 13(143) \pmod{19} = 16$$

signature: $(r, s) = (15, 16)$

iii) verification

1. calculate weights: $w = s^{-1} \pmod{19} = 6$

$$u_1 = H(M)w = 8 \cdot 6 = 48 \pmod{19} = 10$$

$$u_2 = rw = 15 \cdot 6 = 90 \pmod{19} = 14$$

2. compute R_{ind} : $*X = u_1G + u_2G = 10G + 14G = (15, 20)$

3. Check: The x-coordinate (15) matches

 r . valid

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Ans: i) Cone Properties

- Pre-image-Resistance: Given h , Impossible to find m where $H(m)=h$
- Second-Pre-Image Resistance: Given m_1 , impossible to find m_2 that hashes to the same value
- Collision Resistance: Impossible to find any two inputs that hash to the same value.

ii) Output Length & Security

- Brute force: Length n dictates attack difficulty
- Birthday Paradox: Finding collisions takes $2^{n/2}$ operations.

iii) Application:

- Digital signature: Signs the hash of a message for speed and integrity.
- Blockchain: Hashes link blocks together any change breaks the chain.
- Proof of work: Miners must find a hash meeting specific criteria to validate blocks.

22.

Ans: finite fields (GF), A Galois field is a finite set where $+$, $-$, $\times$ and $\div$ are always defined.

Types.

- $GF(P)$ (Prime fields)
- $GF(2^n)$ (Binary Extension field)

Cryptographic use:

- AES ($GF(2^8)$): The S-Box uses multiplicative inversion in this field to scramble data.

$GF(2^n)$ is used because addition is a simple, fast bitwise XOR.

ECC: Curve coordinates are elements of $GF(P)$ or $GF(2^n)$.

Importance:

1. Precision

2. Hardware Speed

23.

Ans: i) Shortest Vector Problem (SVP): A lattice is a grid of points in n -dimensional space. The SVP asks to find the shortest non-zero vector (point closest to the origin) in that lattice.

ii) comparison between Post Quantum and Traditional Quantum.

Feature	Traditional (RSA/ECC)	Lattice-Based Cryptography
Hard Problem	Factoring / Discrete Log	SVP / Learning with Errors (LWE)
Short's Algorithms	Broken: Can solve these in polynomial time.	Resistant: No known efficient quantum algorithm exists.
Key Size	Small / efficient	Larger than RSA/ECC but manageable

29.

Ans: Proof: Maximum period of an LFSR

1. State Constraints: An m -bit register has 2^m total bit combinations.

• The zero state: $(0, 0, \dots, 0)$ must be excluded because the XOR sum of zeros is always zero, preventing the sequence from ever changing.

• Non-zero States: This leaves exactly $2^m - 1$ possible states.

2. Algebraic Property: The LFSR's recurrence corresponds to a characteristic polynomial over $\text{GF}(2)$.

3. Conclusion: Since the LFSR visits all available non-zero states before returning to the start, the maximum period T is:

$$T = 2^m - 1$$

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Ans: LWE signature : Quick so.

i) Process :

1. KeyGen : Secret s (private), $t = As + e$ (public)

2. Sign : Create mask y , challenge $c = \text{Hash}(M, Ay)$
and response $z = y + cs$

3. verify = Check if $Az - ct \approx Ay$ and if z is small

ii) Signing & security

Commit : Pick random y compute $w = Ay$

challenge : $c = \text{Hash}(M, w)$

output : (c, z) where $z = y + cs$